

Deriving the Multi-Dose Concentration Formula From a First-Order Differential Equation via an Infinite Geometric Series

Brian Tenneson

M.S., Applied Data Science - Currently Unaffiliated

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email: btenneson2301@baypath.edu

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btenneson2301.substack.com

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Abstract

Beginning with the first-order elimination law $c'[t] = k_1 c[t]$, I construct the single-dose concentration profile, superpose n doses to obtain a finite geometric series, and pass to the limit $n \rightarrow \infty$. The steady-state peak and trough then follow by evaluating a single closed form at the two endpoints of one dosing cycle, and the finite- n expression reproduces the ManyDoseConcentration function of the underlying Mathematica notebooks exactly.

The derivation reduces to one structural observation: a sum of exponential tails offset by equal time steps is a geometric series. Anchoring that sum at the most recent dose makes the closed form, its limit, and both clinical extremes follow in a few lines of algebra. Elimination is written in base two so the half-life is read directly off every formula. Default dosing period: one dose per 24 hours.

Keywords

pharmacokinetics; first-order elimination; half-life; absorption ramp; superposition; geometric series; steady state; accumulation multiplier; peak; trough; fluctuation index; ManyDoseConcentration; closed form; dosing period.

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Symbols and standing conventions

t = time [h]; h = elimination half-life [h]; x = absorption duration [h];
 P = dosing period [h, default 24]; τ = time since current dose finished absorbing.
Base-two convention: elimination is written as $2^{-t/h}$, so after one half-life the factor is $1/2$, after two it is $1/4$, and after m it is 2^{-m} .

1 The Elimination Law and Its Half-Life Parameterisation

A drug is cleared at a rate proportional to the amount present. This first-order law is the entire physical input to the model.

Definition 1.1 - First-order elimination law

$$c'[t] = k_1 c[t], \quad k_1 < 0. \quad [1]$$

Proposition 1.2. Every solution has the form

$$c[t] = k_2 \exp[k_1 t]. \quad [2]$$

Separating variables gives $dc/c = k_1 dt$; integrating gives $\ln|c| = k_1 t + \text{const.}$
Thus $c[t] = k_2 \exp[k_1 t]$.

Proposition 1.3 - Half-life reparameterisation

The half-life h is defined by $c[t+h] = [1/2] c[t]$. Imposing this on [2] gives

$$\exp[k_1 h] = 1/2 \iff k_1 h = -\ln 2 \iff k_1 = -[\ln 2]/h. \quad [3]$$

Substituting [3] into [2] gives

$$c[t] = k_2 * 2^{-t/h}. \quad [4]$$

Remark 1.4. In base two the half-life is manifest; powers of 2 replace powers of e from this point on.

2 The Single-Dose Concentration Profile

An oral dose is not absorbed instantaneously. Over an absorption window of length x the drug enters the blood at a constant rate; afterwards [4] governs its decay.

Definition 2.1 - Single-dose profile, normalised to unit concentration at $t = x$

$$\begin{aligned} c_{\text{single}}[t] &= t/x, & 0 < t \leq x; \\ c_{\text{single}}[t] &= 2^{[-(t-x)/h]}, & t > x. \end{aligned} \quad [5]$$

The profile satisfies $c_{\text{single}}[0] = 0$ and $c_{\text{single}}[x] = 1$ and is continuous at $t=x$.

Definition 2.3 - Finish-of-absorption time

$$t_k = [k-1]P + x. \quad [6]$$

For $t > t_k$ the residual tail of dose k is

$$c_k[t] = 2^{[-(t-t_k)/h]} = 2^{[-(t-x-[k-1]P)/h]}. \quad [7]$$

3 Superposition of n Doses: A Finite Geometric Series

Equation [1] is linear, so contributions add. After n doses, at a time t in the elimination phase of the current cycle,

$$C_n[t] = \sum_{k=1..n} 2^{[-(t-x-[k-1]P)/h]}. \quad [8]$$

Form A - anchored at the first dose:

$$\begin{aligned} C_n[t] &= 2^{[-(t-x)/h]} \sum_{j=0..n-1} [2^{[P/h]}]^j \\ &= 2^{[-(t-x)/h]} [2^{[nP/h]} - 1] / [2^{[P/h]} - 1]. \end{aligned} \quad [9]$$

Here the ratio $2^{[P/h]}$ is greater than 1: measured from the first dose, later terms are larger. Form A is the expression implemented in the notebooks.

3 Superposition of n Doses - continued

Let tau be elapsed time since the current dose finished absorbing:

$$\tau = t - t_n = t - x - [n-1]P, \quad 0 \leq \tau \leq P-x. \quad [10]$$

Re-index [8] by $i=n-k$, cycles into the past. Then Form B is

$$\begin{aligned} C_n[\tau] &= \sum_{i=0}^{n-1} 2^{-[\tau+iP]/h} \\ &= 2^{-\tau/h} \sum_{i=0}^{n-1} r^i, \quad r := 2^{-P/h}, \quad 0 < r < 1. \end{aligned} \quad [11]$$

$$\begin{aligned} C_n[\tau] &= 2^{-\tau/h} [1-r^n]/[1-r] \\ &= 2^{-\tau/h} [1-2^{-nP/h}]/[1-2^{-P/h}]. \end{aligned} \quad [12]$$

The two forms agree after substituting $t=[n-1]P+x+\tau$ into [9]. Only Form B has a finite $n \rightarrow \infty$ limit because it holds tau fixed inside the current cycle.

Mixing the prefactor of one form with the ratio of the other is the algebraic trap.

4 The Steady-State Limit

Since $0 < r < 1$, $r^n = 2^{-nP/h} \rightarrow 0$. Therefore

$$\begin{aligned} C_{\infty}[\tau] &= 2^{-\tau/h} / [1-2^{-P/h}] \\ &= 2^{[P-\tau]/h} / [2^{P/h}-1]. \end{aligned} \quad [13]$$

Written as the tails of all past doses,

$$\begin{aligned} C_{\infty}[\tau] &= 2^{-\tau/h} + 2^{-[\tau+P]/h} + 2^{-[\tau+2P]/h} + \dots \\ &= 2^{-\tau/h} / [1-2^{-P/h}]. \end{aligned} \quad [14]$$

The accumulation multiplier is $[1-2^{-P/h}]^{-1} = 2^{P/h} / [2^{P/h}-1]$.

If $h \gg P$ there is strong accumulation; if $h \ll P$ the multiplier tends to 1.

5 Peak, Trough, and the Fluctuation Index

Formula [13] decreases monotonically in tau, so the steady-state extremes occur at the endpoints of tau in [0,P-x].

Theorem 5.1 - Steady-state peak, at tau = 0

$$\begin{aligned} \text{peak_ss} &= C_{\text{inf}}[0] = 1/[1-2^{-P/h}] \\ &= 2^{P/h}/[2^{P/h}-1] = 1 + 1/[2^{P/h}-1]. \end{aligned} \quad [15]$$

The 1 is the dose just absorbed; the remaining term is everything left from prior doses.

Theorem 5.2 - Steady-state trough, at tau = P-x

$$\begin{aligned} \text{trough_ss} &= C_{\text{inf}}[P-x] = 2^{-[P-x]/h}/[1-2^{-P/h}] \\ &= 2^{x/h}/[2^{P/h}-1]. \end{aligned} \quad [16]$$

Corollary 5.3 - Fluctuation index

$$\text{peak_ss} / \text{trough_ss} = 2^{[P-x]/h}. \quad [17]$$

The swing depends only on how many half-lives fit into the gap P-x between the end of absorption and the next dose. Larger x or smaller P flattens the profile; a short half-life sharpens it.

Proposition 5.4 - Agreement with notebook definitions, P = 24

$$\begin{aligned} \text{trough_min}[x, h] &= 2^{x/h}/[2^{24/h}-1]. \quad [18] \\ \text{peak_max}[x, h] &= 1/[2^{24/h}-1] + [[12+x-12 \text{ ceil}[x/24]] \text{ ceil}[x/24]]/x. \quad [19] \end{aligned}$$

For $0 < x \leq 24$, $\text{ceil}[x/24]=1$, so the second term in [19] reduces to 1 and [19] becomes exactly [15]. The ceiling expression generalises $x > P$.

6 Recovering the ManyDoseConcentration Function

For finitely many doses the notebooks evaluate Form A, [9], using ceilings to identify the current cycle. With $P=24$, set

$$n = \text{ceil}[t/24], \quad m = \text{ceil}[(t-x)/24]. \quad [20]$$

Thus n counts doses administered and m counts those that have finished absorbing.

Theorem 6.2 - Complete finite-n profile

For $0 < t \leq x$:

$$C[t;x,h] = [12+t-12n] n/x.$$

For $t > x$:

$$C[t;x,h] = 2^{[x-t]/h} [2^{[24m/h]-1}/[2^{[24/h]-1}] [n-m] - [-12-t+12n+12m]/x. \quad [21]$$

The first bracket of the lower branch is Form A with nP replaced by $24m$: the finite geometric series summing tails of the m doses that have completed absorption. The second bracket is the absorption-ramp correction and vanishes whenever $n=m$. Letting $m \rightarrow \infty$ with τ fixed returns [13].

7 Summary and Conclusion

The derivation reduces to one observation: a sum of exponential tails offset by equal time steps is a geometric series. Anchoring the sum at the most recent dose makes the closed form, its limit, and both clinical extremes follow in a few lines of algebra. The base-two convention makes half-life visible at every stage. The finite- n ceiling form is Form A of the same series with the current cycle identified by [20].

Step 1: $c' = k_1 c \rightarrow c[t] = k_2 2^{-t/h}$.
Step 2: ramp t/x on $[0,x]$, then $2^{-[t-x]/h}$.
Step 3: $C_n[\tau] = 2^{-[\tau/h]} [1-r^n]/[1-r]$, $r=2^{-P/h}$.
Step 4: $C_{\infty}[\tau] = 2^{-[\tau/h]}/[1-2^{-P/h}]$.
Step 5: peak, trough, and fluctuation follow from the endpoints.
Step 6: finite- n form is ManyDoseConcentration, with the cycle ceilings.

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This document was generated using Fable from Mathematica notebooks created by the author. Source material: drugs from scratch Mathematica notebooks, Levels 1-8, including the Cymbalta application, together with the Python translation ManyDoseConcentrationdrugs2.ipynb. All mathematical claims, interpretations, and final decisions remain the responsibility of the author.

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Core result

```
C_inf[tau] = 2^[-tau/h]/[1-2^[-P/h]]
peak_ss = 1/[1-2^[-P/h]]
trough_ss = 2^[x/h]/[2^[P/h]-1]
peak_ss / trough_ss = 2^[P-x]/h]
```

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